GENESIS OS / RELATIONSHIP SCORES

Missing Layers Engineering Blueprint

Detailed data-capture, labeling, tracking, packet, and handoff specification for everything the workbook does not fully capture at runtime.

Purpose
This document shows exactly what runtime emotional and behavioral data is still missing, when to collect it, how to collect it, how to label and store it, when to send it to Genesis OS, and what the machine should infer from it.

Authority rule
The workbook remains the scoring source of truth for deterministic assessment logic. This blueprint governs runtime telemetry, missing observational data, Human Emotion Genome labeling, packet structure, and later ML intake.

- Workbook basis: 17 sheets, including AnswerBank (310 scored answer rows), AnswerPairQuantum (48,205 pair rows), and polarity v10 layers.
- This blueprint exists because the workbook is excellent at static scored assessment and pair math, but real relationship forecasting also needs event, state, and context data gathered over time.
- Everything here must respect update lanes: slow identity, medium pattern, fast state, and event truth.

1. What the workbook already does versus what is still missing

The workbook already tracks static, deterministic assessment and pair logic extremely well. The missing layers are not random extras; they are the runtime and observational signals needed to make Genesis OS behave like a live intelligence system instead of a one-time questionnaire.

Layer	Already covered in workbook	Still missing outside workbook	Why it matters
Assessment scoring	AnswerBank answer vectors, readiness norms, penalties, archetype/shadow labels	Real-time behavior after onboarding	The engine needs to see whether people live like their answers.
Pair compatibility	AnswerPairQuantum, matrices, overlays, harness	Observed pursuit / withdrawal rhythm, repair quality, silence behavior	Two people can score well on paper and still collapse in motion.
Chemistry	Polarity v10 axes and pair engine	Live attraction drift, desire suppression, strain after conflict	Chemistry changes under stress.
Outcome learning	FeedbackData starter fields	Longitudinal outcome telemetry and confidence history	ML cannot learn from vibes; it needs consistent labels and timestamps.
Narrative intelligence	Some style labels in AnswerBank	Full Human Emotion Genome from words, tone, pacing, and events	The machine needs governed narrative signals, not just scores.

Workbook registry snapshot

Sheet	Rows	Cols	Role
README_v10	11	1	Static / pair / reference
README	9	5	Static / pair / reference
Schema	18	4	Static / pair / reference
FeedbackData	17	4	Runtime support
ArchetypeMatrix	10	11	Static / pair / reference
ShadowMatrix	6	7	Static / pair / reference
AnswerBank	310	35	Static / pair / reference
ZodiacLookup	576	7	Static / pair / reference
NumerologyLookup	66	6	Static / pair / reference

Sheet	Rows	Cols	Role
AnswerPairQuantum	48205	28	Static / pair / reference
MatchHarness	84	14	Runtime support
Polarity_Axes_v10	10	6	Static / pair / reference
Polarity_RoleSeeds_v10	5	24	Static / pair / reference
Polarity_QuestionMap_v10	32	7	Static / pair / reference
Rewrite_Rules_v10	11	3	Static / pair / reference
Polarity_AnswerBank_v10	160	31	Static / pair / reference
Polarity_PairEngine_v10	22	9	Static / pair / reference

2. Core design rule for missing data

- If a signal happens once and matters later, store it as an event.
- If a signal is a current condition, store it as fast state with decay rules.
- If a signal repeats and becomes stable, roll it into a medium-pattern feature with confidence.
- If a signal changes identity-level assumptions, require multiple observations and governance before it can influence slow identity.
- Never let one dramatic conversation rewrite a profile.

3. The missing data map

This is the implementation heart of the blueprint: what is missing, when to get it, how to get it, how to label it, where to store it, when to send it to Genesis OS later, and what it should tell the system.

Missing layer	When to get it	How to get it	Label / event code	Store in	When to send	What it tells Genesis OS
Daily emotional state	Daily check-in, Sanctuary open, or morning reminder	Quick self-report slider + 1-2 tags + free text optional	state_daily / E_STATE_DAILY	Fast state table + event log	Send immediately; aggregate nightly	Current emotional weather, regulation load, baseline drift risk
Post-interaction reflection	Within 30-90 min after date, call, or heavy chat	Prompted reflection form + optional narrative	interaction_post / E_INTERACTION_POST	Interaction table + genome note	Send after submit; roll into 7-day window	Whether connection felt safe, energizing, draining, confusing,

Missing layer	When to get it	How to get it	Label / event code	Store in	When to send	What it tells Genesis OS
						pressured, or reciprocal
Silence response	When reply gap crosses configured threshold	Passive detection + optional self-report	silence_window / E_SILENCE_WINDOW	Event log + pair rhythm metric	Send when threshold hit and when resolved	Whether user spirals, stays regulated, pursues, withdraws, or reframes
Conflict event	At first explicit disagreement or conflict-tagged journal/chat	Conflict form + AI classification if consented	conflict_start / E_CONFLICT_START	Conflict episode object	Send on open, update on close	Conflict style under pressure and intensity of activation
Repair attempt	After apology, reconnection, bid, or follow-up	Structured repair form + message telemetry	repair_attempt / E_REPAIR_ATTEMPT	Repair events + outcome status	Send each attempt and final result	Whether user repairs, delays, deflects, over-explains, stonewalls, or restores safety
Boundary signal	When user says no, asks for space, or requests clarity	Message detection + self-tag + coach tool	boundary_signal / E_BOUNDARY_SIGNAL	Boundary event + confidence	Send after capture; review weekly	Boundary clarity, fear level, and whether stated boundaries hold or collapse
Trigger activation	Whenever user reports being activated or AI sees a sharp state change	Trigger form + narrative extraction + optional chat signal	trigger_activation / E_TRIGGER_ACTIVATION	Trigger object with stimulus, body, story	Send immediately; decay after review	What contexts activate attachment, shame, anger, grief, or fantasy loops
Pursuit / withdrawal rhythm	Continuously in message streams and interaction logs	Telemetry from who initiates, who delays, who exits, who returns	pair_rhythm / E_PAIR_RHYTHM	Pair metrics table	Compute nightly and after major episodes	Actual relational dance, not just claimed communication style
Perception-vs-reality gap	After enough observed behavior exists	Compare self-report to event telemetry	perception_gap / E_PERCEPTION_GAP	Derived metric table	Weekly batch or milestone run	How accurate the user is about themselves and where coaching is needed

Missing layer	When to get it	How to get it	Label / event code	Store in	When to send	What it tells Genesis OS
Couple mode health session	Weekly once in official relationship mode	Joint reflection + separate private reflection	couple_health / E_COUPLE_HEALTH	Couple health object	Weekly packet	Shared stability, asymmetry, repair load, intimacy safety
Ex-related activation	When ExsAnonymous or reflection mentions ex repeatedly	Ex module intake + pattern classifier	ex_activation / E_EX_ACTIVATION	Ex module event + genome family	Send immediately, summarize monthly	Unfinished attachment energy still shaping present behavior
Outcome telemetry	7d / 30d / 90d checkpoints	Automated pulse + self-report + relationship status	outcome_checkpoint / E_OUTCOME_CHECKPOINT	Outcome table	Scheduled packet	Whether the system forecasts were directionally right

4. Canonical labels and namespaces

Developers need a rigid naming system. If naming is loose, the machine learning layer becomes junk storage with prettier folder names.

Object type	Prefix	Example	Used for
Event	E_	E_CONFLICT_START	Point-in-time occurrence
State	S_	S_DAILY_REGULATION_LOAD	Current fast state snapshot
Metric	M_	M_PAIR_PURSUIT_ASYMMETRY_14D	Computed rolling measure
Genome family	G_	G_TRIGGER_ABANDONMENT	Human Emotion Genome dimensions
Packet block	P_	P_INTERACTION_SUMMARY	Structured outbound handoff
Consent scope	C_	C_CHAT_ANALYSIS_OPT_IN	Read / save / expose rules
Confidence	CF_	CF_MEDIUM	Evidence strength
Lane	L_	L_FAST	Update speed / governance class

Recommended event catalog

Event code	Lane	Source	Core payload
E_STATE_DAILY	Fast	User self-report	Mood, body load, hope, openness, depletion
E_INTERACTION_POST	Fast	User reflection	How the interaction felt and what changed
E_SILENCE_WINDOW	Fast	Telemetry	Gap threshold hit, chase/pause behavior, resolution
E_CONFLICT_START	Event	User or AI	Conflict opening, topic, intensity
E_CONFLICT_CLOSE	Event	User or AI	Resolved, unresolved, withdrew, ruptured
E_REPAIR_ATTEMPT	Event	User or telemetry	Apology, bid, reassurance, accountability move
E_BOUNDARY_SIGNAL	Event	User or telemetry	Need, limit, pace, access request
E_TRIGGER_ACTIVATION	Fast	User or AI	Activation class, body signal, story loop
E_PAIR_RHYTHM_DAILY	Medium	Computation	Initiation and withdrawal rhythm
E_OUTCOME_CHECKPOINT	Slow	Scheduled workflow	Relationship status and prediction feedback

5. How to label the data

- Every record needs at minimum: user_id, pair_id if applicable, event_code or state_code, source_type, consent_scope, timestamp_utc, local_time_context, confidence_band, and provenance.
- Narrative text must never be stored without linked extracted labels. Save both the raw text and the machine-readable tags.
- Tags should be multi-select but from controlled vocabularies. Free text is useful for context, not for replacing labels.
- Keep label families separate: emotion, trigger, need, boundary, pacing, repair, conflict, attraction, depletion, and outcome.

Required controlled vocabularies

Family	Suggested enum set	Used by
Emotion primary	calm, hopeful, anxious, angry, ashamed, numb, lit_up, drained	User state, interaction reflections, conflict
Trigger class	abandonment, criticism, engulfment, inconsistency, rejection, comparison, ambiguity	Activation and perception gap analysis
Need signal	clarity, reassurance, space, consistency, affection, accountability, pacing	Boundary and coaching logic
Boundary type	time, pace, access, exclusivity, communication, physical, emotional	Boundary events and couple agreements
Repair move	apology, ownership, reassurance, clarification, attunement, follow_through, timeout_then_return	Repair quality scoring
Outcome label	closer, steadier, confused, pressured, disconnected, ruptured, repaired	Feedback and forecast training labels
Attraction drift	stronger, same, weaker, mixed, avoidant_pull, fantasy_only	Chemistry drift modeling
Silence response	regulated_wait, anxious_reach, hard_withdraw, narrative_spiral, protest, reframe	Pursuit/withdrawal fingerprint

6. How to track the data over time

Lane	What belongs here	Update rule	Decay / promotion rule	Can change score directly?
Fast state	Mood, activation, silence stress, conflict intensity	Every event or check-in	Decay over hours/days; summarize into windows	No direct slow-score rewrite
Medium pattern	Repair consistency, pursuit asymmetry, boundary follow-through	Rolling 7d/30d/90d windows	Promote only after repeated evidence	May affect forecasts and coaching
Slow identity	Archetype stability, dominant shadow tendency, durable relational style	Only after strong repeated evidence + governance	Manual or governed model review	Only via controlled update path
Outcome truth	Whether relationships held, ruptured,	Checkpoint schedules	Permanent historical record	Feeds calibration, not instant user

Lane	What belongs here	Update rule	Decay / promotion rule	Can change score directly?
	improved, ghosted, stabilized			punishment

Tracking windows that should exist from day one

Window	Track	Primary use
24h	State volatility, activation count, reassurance demand spikes	Fast-state coaching
7d	Interaction quality average, conflict count, repair count, silence stress average	Short-term forecasting
30d	Pursuit/withdrawal asymmetry, boundary follow-through, attraction drift trend	Relationship health trend
90d	Stability, growth direction, self-awareness gap, forecast calibration	Model training labels and profile reviews

7. What to send to Genesis OS later

Do not stream raw chaos into Genesis OS. Send packetized, governed summaries plus linked evidence.

Packet	What it contains	When to send	Key fields
P_USER_STATE_SNAPSHOT	Current fast state and last 7d averages	After daily check-in, activation, or major interaction	state_primary, regulation_load, depletion, openness, confidence
P_INTERACTION_SUMMARY	Post-date / post-call / post-chat reflection block	After interaction submit	felt_safe, felt_pursued, felt_pressured, reciprocity, attraction_delta, notes_ref
P_CONFLICT_EPISODE	Conflict + repair chain summary	On conflict close	intensity, topic, style_tags, repair_moves, resolution_state
P_PAIR_RHYTHM_ROLLUP	Observed pair dance over window	Nightly / weekly	initiation_ratio, response_latency, withdrawal_count, reentry_count
P_TRIGGER_FINGERPRINT	Activation classes and body/story loop map	On trigger submit and monthly	trigger_class, stimulus, body_signal, story_loop, common_partner_context

Packet	What it contains	When to send	Key fields
P_OUTCOME_TRAINING_LABEL	Checkpoint truth for calibration	7/30/90d	status_label, breakup_flag, clarity_gain, forecast_hit_miss

What Genesis OS should do with each packet

Packet	What it tells Genesis OS
State snapshot	Update fast-state dashboard; decide whether to prompt support, stillness, journaling, or no action.
Interaction summary	Refine reciprocity, safety, pressure, and attraction drift signals for the pair and for the user.
Conflict episode	Update conflict and repair metrics, contradiction checks, and coaching prompts.
Pair rhythm rollup	Compare claimed communication style versus observed cadence; refresh perception-gap metrics.
Trigger fingerprint	Grow the Human Emotion Genome with trigger families, body markers, and recurring narratives.
Outcome training label	Teach calibration pipelines whether prior forecasts, risk flags, and pair scores were directionally useful.

8. Human Emotion Genome mapping

The missing layers are the intake routes for the Human Emotion Genome. The genome should not be a poetic blob. It should be a governed feature family system.

Genome family	Question it answers	Primary sources	Lane
G_REGULATION	How the person returns to baseline under stress	daily state, trigger events, repair timing	medium / slow
G_TRIGGER	What reliably activates fear, shame, anger, or fantasy loops	trigger forms, conflict events, narrative extraction	medium
G_BOUNDARY	How clearly needs and limits are named and held	boundary events, couple agreements, message tags	medium
G_PACING	Whether the person rushes, freezes, or calibrates	interaction summaries, silence windows	medium
G_RECIPROCITY	Whether energy exchange feels mutual or lopsided	post-interaction reflections + telemetry	medium
G_REPAIR	How the person apologizes, returns, attunes, or avoids	conflict and repair episodes	medium / slow

Genome family	Question it answers	Primary sources	Lane
	repair		
G_PERCEPTION_GAP	How self-report differs from observed behavior	assessment versus telemetry comparison	slow
G_ATTRACTION_DRIFT	How desire changes with safety, ambiguity, pressure, and distance	chemistry summaries and pair rhythm	medium
G_EX_IMPRINT	How unresolved past bonds distort present reactions	ExsAnonymous and trigger loops	slow

9. Developer implementation pattern

Build step	Deliverable	What it means
Step 1	Create canonical tables	user_state_daily, interaction_reflection, conflict_episode, repair_event, boundary_event, trigger_event, pair_rhythm_daily, outcome_checkpoint, genome_feature, packet_outbox
Step 2	Create enums / controlled labels	Event codes, lanes, confidence bands, source types, emotion/trigger/boundary/repair enums
Step 3	Build capture surfaces	daily check-in UI, post-interaction form, conflict close form, trigger form, couple health pulse
Step 4	Build telemetry collectors	message latency, initiation count, return-after-withdrawal, silence threshold triggers
Step 5	Build packet builder	Transform raw records into governed packet blocks for Genesis OS ingestion
Step 6	Build genome extractor	Map events and narratives into feature families with confidence and provenance
Step 7	Build weekly summarizers	7d / 30d / 90d rollups and perception-gap comparisons
Step 8	Connect to Genesis OS services	State service, pair service, reflection service, forecast service, feature store

Build step	Deliverable	What it means
Step 9	Add QA dashboards	Missing labels, stale states, broken packet rates, consent mismatch flags

10. Build order by phase

Phase	What to build	Why
Phase 1 - Must ship first	Daily state, post-interaction reflection, conflict/repair events, packet builder, outcome checkpoints	This creates the minimum missing-data spine.
Phase 2 - Pair intelligence	Silence windows, pair rhythm metrics, boundary events, perception-gap metrics	This gives the engine eyes instead of just opinions.
Phase 3 - Genome depth	Trigger fingerprints, ex-imprint module, attraction drift, couple health sessions	This deepens the Human Emotion Genome.
Phase 4 - ML readiness	Feature store, quality scoring, training labels, calibration monitors	This turns the data into learning material.

11. Data quality rules

Rule	Meaning	Developer action
No orphan records	Every event must resolve to a user and timestamp	Reject write if missing core identifiers
Consent first	No chat-derived labels without active consent scope	Policy service gate before save
Provenance required	Know whether data came from self-report, telemetry, AI extraction, or coach workflow	Store source_type and evidence_ref
Confidence bands	Low-confidence AI guesses cannot mutate slow identity	Use CF_LOW / CF_MEDIUM / CF_HIGH
Separate raw from inferred	Never overwrite raw user words with model labels	Store both and link them
Decay stale fast state	Yesterday's panic is not today's identity	Expire fast-state influence on schedule

12. Example end-to-end flows

Flow A - Post-date packet

- User completes a short post-date reflection.
- UI stores raw response, controlled labels, and confidence = self-report-high.
- Packet builder emits P_INTERACTION_SUMMARY.
- Genesis OS updates reciprocity, safety, pressure, and attraction drift metrics.
- If repeated confusion / pressure appears, reflection service can suggest a prompt or caution.

Flow B - Silence spiral detection

- Telemetry sees reply gap cross threshold.
- System logs E_SILENCE_WINDOW with passive source.
- User optionally labels their internal response: regulated_wait, anxious_reach, narrative_spiral, or hard_withdraw.
- Pair rhythm summary updates at nightly rollup.
- If asymmetry and anxiety repeat over 30d, perception-gap and pacing metrics rise.

Flow C - Conflict to repair chain

- Conflict begins; log E_CONFLICT_START.
- Both users can privately reflect on topic and intensity.
- Any apology / repair bid logs E_REPAIR_ATTEMPT.
- Conflict closes as repaired, unresolved, or ruptured.
- Packet builder sends P_CONFLICT_EPISODE and updates genome families G_REPAIR and G_REGULATION.

13. The exact question this missing data answers

Missing signal family	What Genesis OS can know after capture
Daily state + activation	How the person feels right now, not just how they scored on onboarding.
Interaction reflections	Whether connection actually feels safe, mutual, pressuring, or draining.
Silence windows + pair rhythm	How the relational dance behaves when one person is not immediately available.
Conflict and repair	Whether the pair can rupture and return instead of just idealizing each other.
Boundary events	Whether stated needs exist in reality or vanish under

Missing signal family	What Genesis OS can know after capture
	pressure.
Trigger fingerprints + ex-imprint	What old pain keeps writing itself into the present.
Outcome checkpoints	Whether the forecasting engine is earning the right to be trusted.

14. Minimum fields every missing-layer record should carry

Field	Type	Purpose
record_id	uuid	Unique durable ID
user_id	uuid	Primary user key
pair_id	uuid nullable	Pair key when signal involves a match or couple
event_code_or_state_code	text enum	Canonical namespace label
source_type	enum	self_report / telemetry / ai_extract / coach / system
consent_scope	enum	What this signal is allowed to read, save, and expose
lane	enum	fast / medium / slow / event
confidence_band	enum	low / medium / high
occurred_at_utc	timestamp	When it happened
captured_at_utc	timestamp	When system stored it
evidence_ref	text/json	Link to raw note, message range, or form
payload_json	json	Structured field block

15. Final implementation doctrine

- The workbook scores the person and the pair at assessment time. The missing layers explain whether life after onboarding confirms, complicates, or contradicts those scores.
- Do not wait for perfect machine learning before collecting this data. Collect clean event truth first.
- Do not ship raw emotional essays into the feature store without labels, provenance, and confidence.

- Do not let companion or missing-layer systems overwrite core score law. They enrich, gate, explain, and eventually calibrate it.
- If developers build only one thing from this document first, build the event spine, packet builder, and outcome checkpoints.